OPPERMANN INTERVALS IN $\mathbb{F}_q[t]$: COUNTEREXAMPLES IN CHARACTERISTIC 2 AND A COMPUTER-ASSISTED THEOREM IN ODD CHARACTERISTIC

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ABSTRACT. For a monic $f \in \mathbb{F}_q[t]$ of degree d the Legendre interval $\{f^2 + s : \deg s \le d\}$ splits into the q *Oppermann sub-intervals* $O_{f,c} = \{f^2 + cf + s : \deg s < d\}$, $c \in \mathbb{F}_q$, each of size q^d; for $q = 2$ these are the two halves, the analogue of $(n^2, n^2 + n)$ and $(n^2 + n, (n + 1)^2)$. We ask whether every $O_{f,c}$ contains an irreducible polynomial. In characteristic 2 the answer is no, and the failures are explained completely by three Galois-theoretic obstructions: (A) for $d = 1$ the piece $c = 0$ consists of squares; (B) for $d = 2$ we compute the exact number of irreducibles in every Oppermann sub-interval, $q^2/4$, 0, $q(q-2)/4$ or $q(q-1)/2$, by letting Frobenius act on the roots of an additive polynomial through $\mathrm{AGL}_2(\mathbb{F}_2) \cong S_4$; it vanishes exactly when $q = 4^j$, $a_1^2 + c = 0$, or $q = 2$, $a_1^2 + c = 1$; (C) for $q = 2$, $d = 4$ the block $\{t^8 + s : \deg s \le 3\}$ contains no irreducible, by the Stickelberger–Swan sign of Frobenius, because $8 \mid \deg$ makes the sign depend only on the tail. In odd characteristic every class of the Hayes group of level d is an Oppermann sub-interval. For $d = 2$ we prove the exact count $\pi = (q^2 - 1 - (q - \eta(-1))\eta(2e))/4 > 0$ for every odd q, via a ternary trace form on $\mathbb{F}_{q^4}$; with (B), this settles $d = 2$ for all q. An exact computation with the Hayes explicit formula shows that all of them contain irreducibles for $q = 3$, $d \le 16$; $q = 5$, $d \le 10$; $q = 7$, $d \le 9$; $q = 9$, $d \le 8$; $q = 11$, $d \le 7$; $q = 13$, $d \le 6$. In characteristic 2 the same computation, for $q \in \{2, 4, 8, 16\}$ up to $q^d \le 2^{26}$, finds no empty Oppermann sub-interval beyond (A)–(C). The Weil bound sits at the square-root barrier for these intervals. For $d = 2$ it gives $\Psi \ge q$, and in the empty classes of (B) this is attained exactly, by fourth powers of linear polynomials. For fixed d and large q, the theorem of Bank, Bary-Soroker and Rosenzweig gives nonemptiness in most cases; we determine which.

1. INTRODUCTION

Oppermann's conjecture [7] asserts that for every $n \ge 2$ both (n^2, n^2+n) and $(n^2+n, (n+1)^2)$ contain a prime. It implies Legendre's conjecture and lies beyond the Riemann hypothesis, which gives primes only in intervals $[x, x + C\sqrt{x}\log x]$. Over $\mathbb{F}_q[t]$, Legendre intervals $I_f = \{f^2 + s : \deg s \le d\}$ (f monic of degree d) were studied in [3]. Since $f^2 + cf$ and $f^2 + ct^d$ differ by a polynomial of degree $< d$, the interval I_f is the disjoint union of the q sub-intervals

$$O_{f,c} = \{f^2 + cf + s : \deg s < d\} = \{f^2 + ct^d + s : \deg s < d\}, \qquad c \in \mathbb{F}_q, \tag{1}$$

each of cardinality $q^d = |f|$, the norm of f, exactly as $(n^2, n^2 + n)$ has length n. For $q = 2$ there are two halves. We call $O_{f,c}$ the *Oppermann sub-intervals* and ask:

Does every $O_{f,c}$ contain an irreducible polynomial?

The answer depends on the characteristic.

Theorem 1.1 (characteristic 2)**.** *Let $q = 2^k$.*

(A) *For $d = 1$ the sub-interval $O_{f,0} = \{t^2 + s_0 : s_0 \in \mathbb{F}_q\}$, which is the same set for all q polynomials f, consists of squares. Every $O_{f,c}$ with $c \ne 0$ contains exactly $q/2$ irreducibles.*

Date: October 13, 2026.

2020 *Mathematics Subject Classification.* Primary 11T55; Secondary 11T06, 11N05, 11Y16.

Key words and phrases. Oppermann's conjecture, function fields, irreducible polynomials, short intervals, Hayes characters, Stickelberger–Swan theorem, explicit formula.

Draft, October 2026. Code and data in the accompanying repository (directories `code/`, `data/`).

(B) *For $d = 2$ and $f = t^2 + a_1 t + a_0$, the number of irreducibles in $O_{f,c}$ depends only on $e = a_1^2 + c$ and equals*

	$e = 0$	$e \neq 0$
k *even*	0	$q^2/4$
k *odd*	$q(q-1)/2$	$q(q-2)/4$

The table has two zero entries: k even with $e = 0$, and $q(q-2)/4$ at $q = 2$ (where $k = 1$ is odd). Hence $O_{f,c}$ is empty if and only if either k is even and $c = a_1^2$, or $q = 2$ and $c = a_1^2 + 1$. For $q = 2$ the block $e = 0$ is not *empty: it contains $q(q-1)/2 = 1$ irreducible, $t^4 + t + 1$. For every $q \in \{2, 4, 16, 64, \dots\}$ there is thus exactly one empty sub-interval as a set: B_0, or B_1 when $q = 2$. It is represented by q^2 pairs (f, c), one c for each of the q^2 polynomials f. For $q \in \{8, 32, 128, \dots\}$ there is none.*

(C) *For $q = 2$ and $d = 4$ the block $Z_4 = \{t^8 + s : \deg s \leq 3\}$ contains no irreducible polynomial. It equals $O_{f,c}$ for the 8 pairs $f = t^4 + a_2 t^2 + a_1 t + a_0$ (that is, $a_3 = 0$), $c = a_2$.*

Theorem 1.2 (odd characteristic, computer-assisted)**.** *Let q be odd. Every Oppermann sub-interval $O_{f,c}$ contains an irreducible polynomial whenever*

$$q = 3,\ d \leq 16; \qquad q = 5,\ d \leq 10; \qquad q = 7,\ d \leq 9;$$
$$q = 9,\ d \leq 8; \qquad q = 11,\ d \leq 7; \qquad q = 13,\ d \leq 6.$$

Equivalently (Lemma 2.1), every interval $\{P + s : \deg s < d\}$ with P monic of degree $2d$ contains an irreducible polynomial in these ranges.

Theorem 1.3 (characteristic 2, computer-assisted)**.** *For $q = 2$, $d \leq 26$; $q = 4$, $d \leq 13$; $q = 8$, $d \leq 7$; $q = 16$, $d \leq 5$, the only empty Oppermann sub-intervals are those of Theorem 1.1.*

Theorem 1.4 (odd characteristic, $d = 2$)**.** *Let q be odd and let η be the quadratic character of $\mathbb{F}_q$, with $\eta(0) = 0$. For $f = t^2 + a_1 t + a_0$ and $c \in \mathbb{F}_q$, put $e = 2a_0 + c - a_1^2/2$. Then*

$$\pi(O_{f,c}) = \frac{q^2 - 1 - (q - \eta(-1))\,\eta(2e)}{4}.$$

In particular, every Oppermann sub-interval with $d = 2$ contains at least $q(q-1)/4$ irreducibles if $q \equiv 1 \pmod 4$, and at least $(q+1)(q-2)/4$ if $q \equiv 3 \pmod 4$. Together with Theorem 1.1(B), this settles $d = 2$ for every q.

For fixed d and large q, the theorem of Bank, Bary-Soroker and Rosenzweig [1] settles the question in most cases (Corollary 6.1). The tables above therefore matter mainly for small q, where that theorem gives nothing.

Throughout, a *sub-interval* means the set $O_{f,c}$. Distinct pairs (f, c) can give the same set; counts of sub-intervals are counts of sets, that is, of Hayes classes.

Section 2 identifies the Oppermann sub-intervals with Hayes classes and explains why characteristic 2 is special: the Oppermann classes are concentrated on prefixes close to squares. Section 3 proves Theorem 1.1, and Section 4 proves Theorem 1.4. Section 5 describes the exact computation behind Theorems 1.2 and 1.3. Section 6 explains why the explicit formula alone cannot go further.

2. Oppermann sub-intervals as Hayes classes

For P monic of degree n write $P = t^n + p_{n-1}t^{n-1} + \cdots + p_0$. Two monic polynomials of degree n lie in the same *Hayes class of level l* if they agree in $p_{n-1}, \dots, p_{n-l}$. Each class is a coset $P + \{s : \deg s < n - l\}$, and the map

$$P \mapsto u^n P(1/u) \bmod u^{l+1} = 1 + p_{n-1}u + \cdots + p_{n-l}u^l$$

identifies the classes with the elements of the multiplicative group $G_l = (1 + u\mathbb{F}_q[u])/(1 + u^{l+1}\mathbb{F}_q[u])$ of order q^l. This map is multiplicative: $\mathrm{class}(PQ) = \mathrm{class}(P)\,\mathrm{class}(Q)$ in G_l. The characters of G_l are the short-interval characters of Hayes [4] and Keating–Rudnick [5]. A

character has *exact level* j if it factors through G_j but not through G_{j-1}. There are $q^j - q^{j-1}$ characters of exact level j, for $1 \le j \le l$. By (1), $O_{f,c}$ is the Hayes class of level d and degree $n = 2d$ containing $f^2 + ct^d$.

Lemma 2.1 (odd q)**.** *If q is odd, every Hayes class of level d in degree $2d$ is an Oppermann sub-interval, and it equals $O_{f,c}$ for exactly q pairs (f, c).*

Proof. Write $f = \sum_{i=0}^{d} a_i t^i$ with $a_d = 1$. For $1 \le j \le d$ the coefficient of t^{2d-j} in f^2 is

$$[f^2]_{2d-j} = \sum_{\substack{i+i'=2d-j \\ 0\le i,i'\le d}} a_i a_{i'} = 2a_{d-j} + \Phi_j, \qquad \Phi_j = \sum_{\substack{i+i'=2d-j \\ d-j<i,i'<d}} a_i a_{i'},$$

where the term $2a_{d-j} = 2a_d a_{d-j}$ comes from $\{i, i'\} = \{d, d-j\}$. The sum Φ_j runs over ordered pairs, so it includes the square $a_{d-j/2}^2$ when j is even. Φ_j depends only on $a_{d-1}, \dots, a_{d-j+1}$; for example $\Phi_1 = 0$, $\Phi_2 = a_{d-1}^2$, $\Phi_3 = 2a_{d-1}a_{d-2}$.

The map $(a_{d-1}, \dots, a_1) \mapsto ([f^2]_{2d-1}, \dots, [f^2]_{d+1})$ is therefore triangular with diagonal entries 2. Since $2 \in \mathbb{F}_q^\times$, it is a bijection $\mathbb{F}_q^{d-1} \to \mathbb{F}_q^{d-1}$: the prefix of a class determines $a_{d-1}, \dots, a_1$ uniquely, by solving $a_{d-j} = ([f^2]_{2d-j} - \Phi_j)/2$ for $j = 1, \dots, d-1$ in turn.

The remaining coefficient of the class is that of t^d in $f^2 + ct^d$, namely $2a_0 + \Phi_d + c$, where Φ_d is already fixed. For each of the q values of a_0 there is exactly one c giving the prescribed value. So every class arises, and from exactly q of the q^{d+1} pairs (f, c). □

So in odd characteristic the Oppermann question at degree $2d$ is the question whether *every* short interval of length $q^d = q^{n/2}$ contains a prime: no class is singled out.

Lemma 2.2 (q even)**.** *Let q be even. The Oppermann sub-intervals are exactly the Hayes classes of level d (in degree $2d$) in which the coefficient of t^j is 0 for every odd j with $d < j < 2d$. The coefficients of t^d and of t^j for even j with $d < j < 2d$ are arbitrary. There are $q^{\lceil d/2 \rceil}$ such classes among the q^d classes of level d, and each of them equals $O_{f,c}$ for exactly $q^{\lfloor d/2 \rfloor + 1}$ pairs (f, c).*

Proof. In characteristic 2, $f^2 = \sum a_i^2 t^{2i}$ has no odd terms, $a \mapsto a^2$ is bijective on $\mathbb{F}_q$, and the coefficient of t^d is shifted by c. This identifies the classes that occur.

For the count, note that $a \mapsto a^2$ is additive. So the map $(f, c) \mapsto \mathrm{prefix}(f^2 + ct^d)$ is a surjective homomorphism of additive groups from $\mathbb{F}_q^{d+1}$, the coefficients $(a_0, \dots, a_{d-1}, c)$, onto $\mathbb{F}_q^{\lceil d/2 \rceil}$. Its fibres therefore all have size $q^{d+1-\lceil d/2 \rceil} = q^{\lfloor d/2 \rfloor + 1}$. □

For example, the empty classes of Theorem 1.1 at $d = 1$, 2 and 4 are represented by q, q^2 and $2^3 = 8$ pairs (f, c) respectively.

Thus in characteristic 2 the Oppermann sub-intervals are the short intervals around the $q^{\lceil d/2 \rceil}$ "near-squares" $f^2 + ct^d$, a set of measure $q^{-\lfloor d/2 \rfloor}$ among all short intervals. Squares and near-squares carry Galois-theoretic structure, and that structure produces the counterexamples below.

3. Characteristic 2: the three obstructions

Throughout this section $q = 2^k$ and $\mathrm{Tr} = \mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_2}$.

3.1. **(A)** $d = 1$**.** Here $f = t + a_0$ and $O_{f,c} = \{t^2 + ct + s_0 : s_0 \in \mathbb{F}_q\}$.

For $c = 0$ the members $t^2 + s_0 = (t + \sqrt{s_0})^2$ are squares.

For $c \neq 0$, substitute $t = cy$: $t^2 + ct + s_0 = c^2(y^2 + y + s_0/c^2)$. This is irreducible if and only if $y^2 + y + s_0/c^2$ has no root in $\mathbb{F}_q$, that is, if and only if $\mathrm{Tr}(s_0/c^2) = 1$. Exactly $q/2$ values of s_0 satisfy this.

3.2. **(B) $d = 2$: an additive polynomial and $\mathrm{AGL}_2(\mathbb{F}_2)$.** For $f = t^2 + a_1 t + a_0$ we have $f^2 = t^4 + a_1^2 t^2 + a_0^2$, so

$$O_{f,c} = B_e := \{t^4 + et^2 + \alpha t + \beta : \ \alpha, \beta \in \mathbb{F}_q\}, \qquad e = a_1^2 + c.$$

By Lemma 2.2 the blocks B_e, $e \in \mathbb{F}_q$, are precisely the Oppermann classes for $d = 2$.

Proposition 3.1. *Let $\alpha \neq 0$. Put $L(t) = t^4 + et^2 + \alpha t$ and $R(y) = y^3 + ey + \alpha$.*

1. *$t^4 + et^2 + \alpha t + \beta$ is irreducible for some β if and only if R has exactly one root in $\mathbb{F}_q$.*
2. *If R has exactly one root in $\mathbb{F}_q$, then $t^4 + et^2 + \alpha t + \beta$ is irreducible for exactly $q/2$ values of $\beta \in \mathbb{F}_q$.*

If $\alpha = 0$, every member $t^4 + et^2 + \beta = (t^2 + \sqrt{e}\, t + \sqrt{\beta})^2$ is a square.

Proof. L is $\mathbb{F}_2$-linear and separable ($L' = \alpha \neq 0$), so $V = \ker L \subset \overline{\mathbb{F}}_q$ is a 2-dimensional $\mathbb{F}_2$-space, and $V \setminus \{0\}$ is the set of roots of $L(t)/t = R(t)$. The Frobenius $\varphi(x) = x^q$ acts on V by some $F \in \mathrm{GL}(V) \cong S_3$. Exactly one of three cases holds: $F = 1$ (R has three roots in $\mathbb{F}_q$), F of order 3 (no root), or F a transvection (exactly one root).

Fix β and a root r of $P_\beta = L + \beta$. The roots of P_β form the coset $r + V$. Since $L(\varphi(r)) = \varphi(\beta) = \beta$, we have $\varphi(r) = r + v$ with $v \in V$, and $\varphi(r + w) = r + (Fw + v)$. So Frobenius acts on the roots through the affine map $T(w) = Fw + v$ of $V \cong \mathbb{F}_2^2$, an element of $\mathrm{AGL}_2(\mathbb{F}_2) \cong S_4$. P_β is irreducible if and only if T is a 4-cycle.

- If $F = 1$, T is a translation and $T^2 = 1$.
- If F has order 3, $F - 1$ is invertible, so T has a fixed point.
- If F is a transvection, $T^2(w) = w + (F+1)v$, so T is a 4-cycle if and only if $(F+1)v \neq 0$, i.e. $v \notin V^F$.

This proves the "only if" part of (1); the "if" part follows from (2), since $q/2 \geq 1$. For (2), let F be a transvection, so $(F+1)V = V^F$ is a line. Replacing r by $r + w$ replaces v by $v + (F+1)w$, so the class $[v] \in V/V^F \cong \mathbb{F}_2$ depends only on β, and $\beta \mapsto [v]$ is additive. Its kernel consists of the β for which some root $r' = r + w$ is fixed by φ, i.e. $\beta \in L(\mathbb{F}_q)$. The map $L : \mathbb{F}_q \to \mathbb{F}_q$ is $\mathbb{F}_2$-linear with kernel $V^F = V \cap \mathbb{F}_q$ of order 2, so $|L(\mathbb{F}_q)| = q/2$ and exactly $q/2$ values of β have $[v] \neq 0$. □

Lemma 3.2. *Let $\alpha \neq 0$. The cubic $R(y) = y^3 + ey + \alpha$ has exactly one root in $\mathbb{F}_q$ if and only if $\mathrm{Tr}(e^3/\alpha^2) \neq \mathrm{Tr}(1)$.*

Proof. The roots y_1, y_2, y_3 of R are distinct (R is separable as above), with $y_1 + y_2 + y_3 = 0$ and $y_1 y_2 y_3 = \alpha$. Put $\rho = y_1/y_3 + y_2/y_1 + y_3/y_2$ and $\rho' = y_2/y_3 + y_1/y_2 + y_3/y_1$. Even permutations of the roots fix ρ and odd ones exchange ρ, ρ'; using $y_i + y_j = y_k$,

$$\rho + \rho' = \frac{y_1 + y_2}{y_3} + \frac{y_2 + y_3}{y_1} + \frac{y_3 + y_1}{y_2} = 1.$$

Hence ρ, ρ' are the roots of $z^2 + z + \delta$ with $\delta = \rho\rho' \in \mathbb{F}_q$, and Frobenius acts on the roots of R by an odd permutation (a transposition: exactly one root in $\mathbb{F}_q$) if and only if $\rho \notin \mathbb{F}_q$, i.e. $\mathrm{Tr}(\delta) = 1$. Expanding,

$$\delta = 3 + \sum_k \frac{y_i y_j}{y_k^2} + \sum_k \frac{y_k^2}{y_i y_j} = 1 + \sum_k \frac{\alpha}{y_k^3} + \frac{1}{\alpha} \sum_k y_k^3.$$

From $y_k^3 = e y_k + \alpha$ we get $\sum_k y_k^3 = \alpha$, so $\delta = \sum_k \alpha/w_k$ with $w_k = y_k^3 = e y_k + \alpha$. Substituting $y = (w + \alpha)/e$ (for $e \neq 0$) into $w = y^3$ shows that the w_k are the roots of $w^3 + \alpha w^2 + (\alpha^2 + e^3)w + \alpha^3$, so $\sum_k 1/w_k = (\alpha^2 + e^3)/\alpha^3$ and

$$\delta = 1 + e^3/\alpha^2.$$

For $e = 0$ this also holds directly ($w_k = \alpha$). So $\mathrm{Tr}(\delta) = 1$ if and only if $\mathrm{Tr}(e^3/\alpha^2) \neq \mathrm{Tr}(1)$. □

Remark 3.3. δ is Berlekamp's characteristic-2 discriminant [2] of R, and R is the cubic resolvent of the quartic; the proof above is self-contained. For $e = 0$ the lemma says: $y^3 = \alpha$ has exactly one root iff k is odd, which is the familiar fact that cubing is bijective on $\mathbb{F}_q$ iff $3 \nmid q - 1$.

Proof of Theorem 1.1(B). By Proposition 3.1, $|B_e \cap \{\text{irreducibles}\}| = \frac{q}{2} \cdot \#\{\alpha \neq 0 : \mathrm{Tr}(e^3/\alpha^2) \neq \mathrm{Tr}(1)\}$. If $e = 0$ the condition reads $\mathrm{Tr}(1) = 1$, i.e. k odd: the count is $\frac{q}{2}(q-1)$ for k odd and 0 for k even. If $e \neq 0$, $\alpha \mapsto e^3/\alpha^2$ permutes $\mathbb{F}_q^\times$, and $\#\{x \neq 0 : \mathrm{Tr}(x) = 1\} = q/2$, $\#\{x \neq 0 : \mathrm{Tr}(x) = 0\} = q/2 - 1$; with $\mathrm{Tr}(1) = k \bmod 2$ this gives $q^2/4$ (k even) and $\frac{q}{2}(\frac{q}{2} - 1)$ (k odd). The count $q(q-2)/4$ vanishes only for $q = 2$. □

The formula was checked by brute force for $q \leq 64$ (script `char2_d2_theorem.py`): criterion of Lemma 3.2, the $q/2$ count of Proposition 3.1(2), and the four block totals.

3.3. **(C) $q = 2$, $d = 4$: the sign of Frobenius.** The Oppermann classes for $q = 2$, $d = 4$ are the four prefixes $(p_7, p_6, p_5, p_4) = (0, x, 0, y)$ (Lemma 2.2). For $x = y = 0$ the class is the *zero block*

$$Z_4 = \{t^8 + a_3 t^3 + a_2 t^2 + a_1 t + a_0\} \subset \mathbb{F}_2[t],$$

equal to $O_{f,c}$ for $f = t^4 + a_2 t^2 + a_1 t + a_0$ and $c = a_2$. Here $a_3 = 0$ is forced, because the coefficient of t^6 in f^2 is a_3^2; this is why there are $2^3 = 8$ pairs rather than 16.

We use the theorem of Stickelberger [8] in the form of Swan [9, Cor. 1]: *if $\tilde{P} \in \mathbb{Z}[t]$ is monic of degree n and its reduction $P \in \mathbb{F}_2[t]$ is squarefree with r irreducible factors, then $r \equiv n \pmod 2$ if and only if* $\mathrm{disc}(\tilde{P}) \equiv 1 \pmod 8$. Equivalently, the Frobenius permutation of the roots of P is even if and only if $\mathrm{disc}(\tilde{P}) \equiv 1 \pmod 8$. An irreducible P of degree 8 has Frobenius an 8-cycle, an odd permutation, and would need $\mathrm{disc}(\tilde{P}) \equiv 5 \pmod 8$.

Lemma 3.4. *Let $\tilde{P} = t^n + \tilde{g}$ with $\tilde{g} \in \mathbb{Z}[t]$, $\deg \tilde{g} < n$, and $8 \mid n$. Then* $\mathrm{disc}(\tilde{P}) \equiv \mathrm{Res}(\tilde{P}, \tilde{g}') \pmod 8$.

Proof. $\mathrm{disc}(\tilde{P}) = (-1)^{n(n-1)/2} \mathrm{Res}(\tilde{P}, \tilde{P}') = \prod_{\tilde{P}(\rho)=0} (n\rho^{n-1} + \tilde{g}'(\rho))$, the sign being $+1$ for $8 \mid n$. Expanding the product, every term containing a factor n is n times a symmetric polynomial with integer coefficients in the roots, hence lies in $8\mathbb{Z}$. □

Proof of Theorem 1.1(C). An irreducible member of Z_4 has no root in $\mathbb{F}_2$, so $a_0 = 1$ and $a_1 + a_2 + a_3 = 1$. This leaves four polynomials. $t^8 + t^2 + 1 = (t^4 + t + 1)^2$ is a square. For the other three choose integer lifts and apply Lemma 3.4:

- $t^8 + t + 1$: $\tilde{g}' = 1$, disc $\equiv 1$;
- $t^8 + t^3 + 1$: $\tilde{g}' = 3t^2$, $\mathrm{Res}(\tilde{P}, 3t^2) = 3^8 \big(\prod \rho\big)^2 = 3^8 \equiv 1$;
- $t^8 + t^3 + t^2 + t + 1$, lifted as $t^8 + (t+1)^3$: $\tilde{g}' = 3(t+1)^2$, and $\mathrm{Res}(\tilde{P}, 3(t+1)^2) = 3^8 \prod (\rho + 1)^2 = 3^8 \tilde{P}(-1)^2$, with $\tilde{P}(-1) = (-1)^8 + 0^3 = 1$, so disc $\equiv 3^8 \equiv 1$.

Suppose one of these three were irreducible over $\mathbb{F}_2$. Then it would be squarefree with $r = 1$, and Stickelberger–Swan would require $\mathrm{disc}(\tilde{P}) \equiv 5 \pmod 8$. That contradicts $\mathrm{disc}(\tilde{P}) \equiv 1$, so none of them is irreducible. The argument does not need squarefreeness to be checked beforehand. □

Remark 3.5. The mechanism is that $8 \mid \deg P$ kills the leading term of P' modulo 8, so the sign of Frobenius is decided by the tail g alone; with $\deg g \leq 3$ the tail is too short to produce the sign -1 on a polynomial with no linear factor. Table 1 shows the zero blocks $Z_d = \{t^{2d} + s : \deg s < d\}$, which are Oppermann classes for every d. For $d = 8$ ($8 \mid 2d$ again) the tail has degree ≤ 7, half of the squarefree members already have the sign -1, and the block contains 10 irreducibles. The parallel with (B) is exact: in (B) the obstruction is the quotient $S_4 \to S_3$ (the resolvent cubic), in (C) the quotient $S_8 \to \{\pm 1\}$ (the sign).

4. Odd characteristic, $d = 2$: a ternary trace form

Throughout this section q is odd and η is the quadratic character of $\mathbb{F}_q$, with $\eta(0) = 0$.

TABLE 1. The zero blocks $Z_d = \{t^{2d} + s : \deg s < d\} \subset \mathbb{F}_2[t]$. Columns: the squarefree members; those with disc $\equiv 5 \pmod 8$, i.e. Frobenius odd (integer lift with coefficients $0, 1$); those among them with no root in $\mathbb{F}_2$; and the irreducibles. For $d = 4$ the two members with disc $\equiv 5$ are t^8+t^3+t and $t^8+t^3+t^2+1$, which have the roots 0 and 1. Stickelberger–Swan holds on every squarefree member (script `swan_parity.py`).

d	$\lvert Z_d\rvert$	squarefree	disc $\equiv 5$	disc $\equiv 5$, no root	irreducible
2	4	2	2	1	1
3	8	4	3	1	1
4	16	8	2	**0**	**0**
5	32	16	10	4	4
6	64	32	20	8	6
7	128	64	42	17	9
8	256	128	64	24	10
9	512	256	130	61	28
10	1024	512	280	120	49

4.1. **Reduction to one-parameter blocks.** The level-2 classes in degree 4 are $C_{p_3,p_2} = \{t^4 + p_3t^3 + p_2t^2 + \alpha t + \beta : \alpha, \beta \in \mathbb{F}_q\}$. The translation $t \mapsto t - p_3/4$ is defined because 4 is invertible (in characteristic 3 it is $t \mapsto t - p_3$). It preserves irreducibility and maps C_{p_3,p_2} bijectively onto

$$B_e = \{t^4 + et^2 + \alpha t + \beta : \alpha, \beta \in \mathbb{F}_q\}, \qquad e = p_2 - \tfrac{3}{8}p_3^2.$$

For $O_{f,c}$ with $f = t^2 + a_1t + a_0$ we have $p_3 = 2a_1$ and $p_2 = a_1^2 + 2a_0 + c$, so $e = 2a_0 + c - a_1^2/2$.

4.2. **The count.**

Proposition 4.1. *For every $e \in \mathbb{F}_q$, the number of irreducibles in B_e is*

$$\pi(B_e) = \frac{q^2 - 1 - (q - \eta(-1))\eta(2e)}{4}.$$

Proof. A monic irreducible quartic $P \in B_e$ is the minimal polynomial of exactly four elements $x \in \mathbb{F}_{q^4}$ of degree 4, namely its roots. Write $\mathrm{Tr} = \mathrm{Tr}_{\mathbb{F}_{q^4}/\mathbb{F}_q}$. The coefficient conditions on P translate into conditions on x: $p_3 = -\mathrm{Tr}(x)$, and when $\mathrm{Tr}\, x = 0$,

$$p_2 = e_2(x, x^q, x^{q^2}, x^{q^3}) = \tfrac{1}{2}((\mathrm{Tr}\, x)^2 - \mathrm{Tr}(x^2)) = -\tfrac{1}{2}\mathrm{Tr}(x^2).$$

Hence $4\pi(B_e) = N - M$, where

$$N = \#\{x \in W : Q(x) = -2e\}, \qquad W = \ker \mathrm{Tr}, \quad Q(x) = \mathrm{Tr}(x^2),$$

and M counts the elements of W with $Q(x) = -2e$ that have degree < 4 over $\mathbb{F}_q$, i.e. that lie in $\mathbb{F}_{q^2}$.

The form Q. The bilinear form $(x, y) \mapsto \mathrm{Tr}(xy)$ on $\mathbb{F}_{q^4}$ is nondegenerate. Its Gram determinant on a basis $\omega_1, \dots, \omega_4$ is $D = \det(\omega_i^{q^j})^2$. Frobenius permutes the columns of the matrix $(\omega_i^{q^j})$ by a 4-cycle, an odd permutation, so $\det(\omega_i^{q^j})$ is not fixed by Frobenius. It therefore lies outside $\mathbb{F}_q$ while its square D lies in $\mathbb{F}_q$, so D is a nonsquare in $\mathbb{F}_q$.

Since $\mathrm{Tr}(1 \cdot x) = \mathrm{Tr}\, x$, the line $\mathbb{F}_q$ is orthogonal to W, and $\mathbb{F}_{q^4} = \mathbb{F}_q \perp W$. The Gram determinant of $\mathbb{F}_q$ is $\mathrm{Tr}(1) = 4$, a square. So $Q|_W$ is a nondegenerate ternary form whose determinant Δ is a nonsquare.

Counting N. For a nondegenerate quadratic form in 3 variables over $\mathbb{F}_q$ with determinant Δ, the number of solutions of $Q(x) = b$ is $q^2 + q\,\eta(-b\Delta)$ [6, Thm. 6.27]; for $b = 0$ this is q^2. With $b = -2e$ and $\eta(\Delta) = -1$ we get $N = q^2 - q\,\eta(2e)$, also for $e = 0$.

Counting M. An element of $\mathbb{F}_{q^2}$ has $\operatorname{Tr} x = 2\operatorname{Tr}_{\mathbb{F}_{q^2}/\mathbb{F}_q}(x)$, so it lies in W if and only if $x^q = -x$. Such x form the line $\delta\mathbb{F}_q$, where $\delta^2 = \nu$ is a nonsquare of $\mathbb{F}_q$. For these x, $x^2 \in \mathbb{F}_q$ and $Q(x) = 2(x^2 + x^{2q}) = 4x^2$, so the condition is $x^2 = -e/2$. The squares of elements $s\delta$ are 0 and the nonsquares of $\mathbb{F}_q$, each nonsquare attained twice. Hence $M = 1$ if $e = 0$, and $M = 1 - \eta(-e/2)$ if $e \neq 0$; in both cases $M = 1 - \eta(-2e)$.

Therefore

$$4\pi(B_e) = q^2 - q\,\eta(2e) - 1 + \eta(-2e) = q^2 - 1 - (q - \eta(-1))\eta(2e). \qquad \square$$

Proof of Theorem 1.4. Combine the reduction with Proposition 4.1. The minimum over e is attained at $\eta(2e) = 1$. It equals $(q^2 - 1 - (q-1))/4 = q(q-1)/4$ when $\eta(-1) = 1$, i.e. $q \equiv 1 \pmod 4$, and $(q^2 - 1 - (q+1))/4 = (q+1)(q-2)/4$ when $q \equiv 3 \pmod 4$. Both are positive for every odd q. $\square$

Remark 4.2. The values agree with a brute-force count of all q^2 members of every block B_e for $q \in \{3, 5, 7, 9, 11, 13, 25, 27, 49, 81\}$, and for $q \leq 13$ also of every class C_{p_3,p_2} (script `odd_d2_formula.py`). They also agree with the minima of Table 2. Characteristic 3 needs no special treatment.

The contrast with characteristic 2 is instructive. There the translation $t \mapsto t + a$ cannot remove p_3, since $4 = 0$. The Oppermann blocks are exactly the classes with $p_3 = 0$, on which $t^4 + et^2 + \alpha t$ is *additive*, and this additivity produces the obstruction of Proposition 3.1. In odd characteristic the trace form $\operatorname{Tr}(x^2)$ has the same shape for every e, and the count never drops below about $q^2/4 - q/4$.

5. The exact computation

5.1. **Method.** Here Λ is the von Mangoldt function: $\Lambda(F) = \deg \varpi$ if $F = \varpi^k$ for a monic irreducible ϖ and some $k \geq 1$, and $\Lambda(F) = 0$ otherwise. For a class x of level l and degree n, let $\Psi(x) = \sum_{F \in x} \Lambda(F)$. The explicit formula [4, 5] reads

$$\Psi(x) = q^{n-l} + q^{-l} \sum_{\chi \neq 1} \overline{\chi}(x)\, \psi_n(\chi), \qquad u\frac{L'(u,\chi)}{L(u,\chi)} = \sum_{m \geq 1} \psi_m(\chi) u^m, \tag{2}$$

where for $\chi \neq 1$ the L-function $L(u,\chi) = \sum_F \chi(F) u^{\deg F}$ is a polynomial of degree $\leq l - 1$. Its coefficients are the Fourier transforms on G_l of the class distributions of the monic polynomials of degree $< l$; $\psi_n(\chi)$ follows by the Newton recursion, and one inverse FFT on G_l returns $\Psi(x)$ for all q^l classes at once. This is the algorithm of [3], run here at level $l = d$ and degree $n = 2d$. The number of irreducibles in x is

$$\pi(x) = \frac{1}{n}\big(\Psi(x) - E(x)\big), \qquad E(x) = \sum_{k \geq 2,\ k \mid n} \ \sum_{\deg \varpi = n/k,\ \varpi^k \in x} \frac{n}{k}.$$

The proper prime powers in $E(x)$ are enumerated directly (Ben-Or test in degree $\leq d$) and placed by the rule $\operatorname{class}(\varpi^k) = \operatorname{class}(\varpi)^k$ in G_l. Each run checks that every $\pi(x)$ is an integer and that $\sum_x (n\pi(x) + E(x)) = q^n$, and reports the largest distance of the floating-point $\Psi(x)$ from an integer (below 10^{-7} in all runs). As an independent check, a brute-force census that tests every polynomial of degree $2d$ for irreducibility (`verify_pointwise_oppermann.py`) agrees with the FFT counts class by class for all $q^{2d} \leq 2^{22}$ (`compare_bruteforce.py`), and in minimum, mean and empty classes for all $q^{2d} \leq 2^{26}$.

5.2. **Results.** Table 2 lists, for odd q, the minimum of $\pi(x)$ over all classes (all of which are Oppermann sub-intervals by Lemma 2.1) and its ratio to the expected value $q^d/2d$. Table 3 gives the same for the Oppermann classes in characteristic 2. The full output, all degrees, is in the file
`data/hayes_oppermann_census.txt`.

Proof of Theorems 1.2 and 1.3. Exact computation as described above; the minima are in Tables 2 and 3, and the empty classes in characteristic 2 occur exactly at

$$(q, d) \in \{(2,1), (2,2), (2,4), (4,1), (4,2), (8,1), (16,1), (16,2)\},$$

one class each, which are the classes of Theorem 1.1. □

The normalised minimum $\min_x \pi(x)/(q^d/2d)$ tends to 1 in every column: the worst Oppermann sub-interval is only slightly below average, far from the Weil bound discussed next.

TABLE 2. Odd q: irreducible counts π over all q^d Hayes classes of level d in degree $2d$ (all are Oppermann sub-intervals). Last four degrees for each q; all smaller degrees are in the data file, with minimum ≥ 1 throughout.

q	d	classes	$\min \pi$	mean π	$\min \pi/(q^d/2d)$
3	13	1 594 323	60 556	61 320.1	0.9875
	14	4 782 969	169 300	170 820.3	0.9911
	15	14 348 907	476 000	478 296.9	0.9952
	16	43 046 721	1 341 030	1 345 210.0	0.9969
5	7	78 125	5 418	5 580.3	0.9709
	8	390 625	23 951	24 414.0	0.9810
	9	1 953 125	107 410	108 506.9	0.9899
	10	9 765 625	486 042	488 281.2	0.9954
7	6	117 649	9 500	9 804.0	0.9690
	7	823 543	57 981	58 824.4	0.9857
	8	5 764 801	358 343	360 300.0	0.9946
	9	40 353 607	2 235 915	2 241 867.0	0.9973
9	5	59 049	5 680	5 904.8	0.9619
	6	531 441	43 728	44 286.7	0.9874
	7	4 782 969	339 993	341 640.6	0.9952
	8	43 046 721	2 684 630	2 690 420.0	0.9978
11	4	14 641	1 735	1 830.0	0.9480
	5	161 051	15 804	16 105.0	0.9813
	6	1 771 561	146 403	147 630.0	0.9917
	7	19 487 171	1 388 036	1 391 940.7	0.9972
13	3	2 197	349	366.0	0.9531
	4	28 561	3 450	3 570.0	0.9664
	5	371 293	36 672	37 129.2	0.9877
	6	4 826 809	400 104	402 234.0	0.9947

TABLE 3. Even q: irreducible counts over the $q^{\lceil d/2 \rceil}$ Oppermann classes (Lemma 2.2). Last four degrees for each q; the only empty classes in the whole range are those of Theorem 1.1.

q	d	classes	$\min \pi$	mean π	$\min \pi/(q^d/2d)$
2	23	4 096	181 349	182 329.3	0.9945
	24	4 096	347 822	349 412.9	0.9951
	25	8 192	669 169	671 025.2	0.9971
	26	8 192	1 287 576	1 290 408.9	0.9977
4	10	1 024	51 908	52 350.0	0.9901
	11	4 096	189 054	190 603.5	0.9916
	12	4 096	695 868	698 820.5	0.9954
	13	16 384	2 575 566	2 580 963.0	0.9979
8	4	64	440	490.0	0.8594
	5	512	3 204	3 276.0	0.9778
	6	512	21 616	21 840.0	0.9895
	7	4 096	149 148	149 796.0	0.9957
16	2	16	**0**	60.0	**0**
	3	256	600	680.0	0.8789
	4	256	7 616	8 160.0	0.9297
	5	4 096	103 680	104 856.0	0.9888

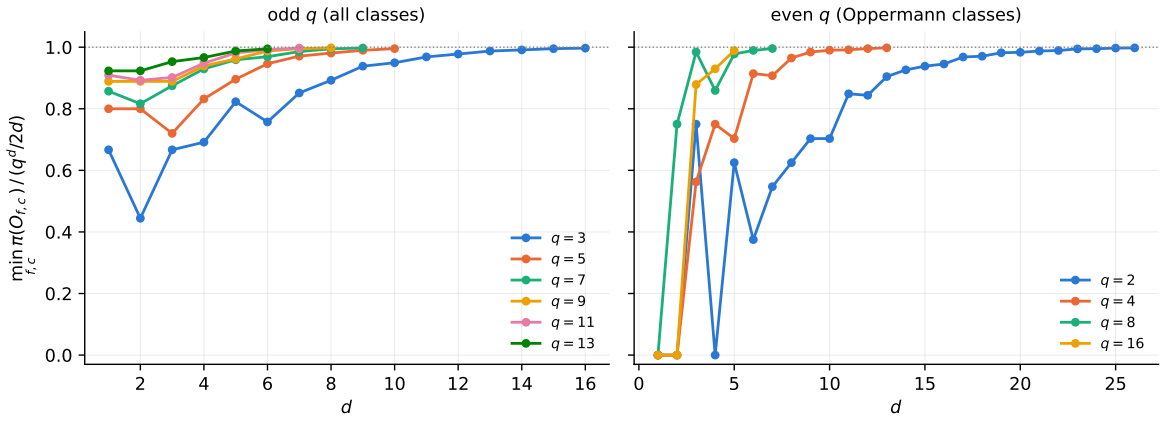


FIGURE 1. Normalised minimum $\min_{f,c} \pi(O_{f,c})/(q^d/2d)$ against d. Left: odd q. Right: even q; the points at 0 are the empty sub-intervals of Theorem 1.1.

6. THE SQUARE-ROOT BARRIER

In (2) with $n = 2d$ and $l = d$ the main term is q^d.

Let χ have exact level j. The monic polynomials of degree $\geq j$ are equidistributed over G_j, so $\deg L(u, \chi) \leq j - 1$. By Weil, every inverse zero γ of $L(u, \chi)$ has $|\gamma| \in \{1, \sqrt{q}\}$, hence $|\psi_{2d}(\chi)| \leq (j-1)q^d$. Characters of level 1 have $L = 1$ and contribute nothing. Multiplying by the factor q^{-d} in (2), which cancels the q^d, and summing over the $q^j - q^{j-1}$ characters of each exact level j gives

$$|\Psi(x) - q^d| \leq \sum_{j=2}^{d} (j-1)(q^j - q^{j-1}) = (d-1)q^d - \sum_{j=1}^{d-1} q^j. \tag{3}$$

For $d \geq 3$ the right-hand side exceeds q^d for every q, so (3) does not even give $\Psi(x) > 0$. This is the exact analogue of the situation over $\mathbb{Z}$: Oppermann's intervals have length $\sqrt{x}$, and the Riemann hypothesis falls short by a factor $\log x$. No degree threshold beyond which $O_{f,c} \neq \emptyset$ follows from Weil's theorem.

For $d = 2$, (3) reads $|\Psi(x) - q^2| \leq q^2 - q$, so $\Psi(x) \geq q > 0$ for every class. This still does not produce an irreducible, because Ψ also counts prime powers. Theorem 1.1(B) shows the lower bound is sharp. Take $q = 4^j$ and the empty class $B_0 = \{t^4 + \alpha t + \beta\}$. Its proper prime powers are the q fourth powers $(t + c_0)^4 = t^4 + c_0^4$, each with $\Lambda = 1$; the squares $(t^2 + b)^2$ do not count, since $t^2 + b$ is itself a square. Hence

$$\Psi(B_0) = 4\pi(B_0) + E(B_0) = 0 + q = q.$$

So equality holds in (3), which forces $\overline{\chi}(B_0)\psi_4(\chi) = -q^2$ for all $q^2 - q$ characters of exact level 2. The whole of the Weil lower bound is taken up by the fourth powers.

The same happens for $q = 2$ on B_1, where $\Psi(B_1) = E(B_1) = 2$ comes from $(t^2+t+1)^2$. The values $\Psi(B_0) = q$ for $q = 4, 16$ and $\Psi(B_1) = 2$ for $q = 2$ were confirmed with the FFT code.

Results on the variance of Ψ over Legendre intervals (an average over f) bound the number of intervals that deviate by a given amount; they do not exclude a single empty interval.

For fixed degree and $q \to \infty$, Bank, Bary-Soroker and Rosenzweig [1] prove the following. For monic $P \in \mathbb{F}_q[t]$ of degree k and $1 \leq m < k$, the number N of irreducibles in $\{P+h : \deg h \leq m\}$ satisfies

$$|N - q^{m+1}/k| \leq c(k)\, q^{m+1/2},$$

provided that $m \geq 2$ if $\gcd(q, k(k-1)) > 1$, and $m \geq 3$ if q is even and $\deg P' \leq 1$. They also note that the estimate fails outside this range.

Our blocks $B_e = \{t^4 + et^2 + h : \deg h \leq 1\}$ have $k = 4$ and $m = 1$, and in characteristic 2 the centre $P = t^4 + et^2$ has $P' = 0$, so both conditions exclude them. Theorem 1.1(B) gives the exact count at this excluded point and shows how the asymptotic fails:

- for $q = 4^j$ the block $e = 0$ contains no irreducible, while the expected count is $q^2/4$;
- for $q = 2^{2j+1}$ the block $e = 0$ contains $q(q-1)/2$, about twice the expected count.

The blocks $e \neq 0$, with $q^2/4$ and $q(q-2)/4$ irreducibles, do agree with the asymptotic. The obstruction comes from the additive structure of $t^4 + et^2 + \alpha t$ (Proposition 3.1). For $m = 1$, $k = 4$ and q even, the conditions of [1] therefore cannot be dropped.

Theorem 1.1(C) has $q = 2$ fixed, so it lies outside the $q \to \infty$ regime altogether.

In the other direction, [1] gives nonemptiness for large q in most cases. An Oppermann sub-interval $O_{f,c} = \{f^2 + ct^d + h : \deg h \leq d - 1\}$ has $k = 2d$ and $m = d - 1$. Since $N \geq q^d/2d - c(2d)q^{d-1/2}$, the sub-interval is nonempty as soon as $q > (2d\, c(2d))^2$, provided the conditions of [1] hold. We do not make the constant $c(2d)$ explicit.

Corollary 6.1 (of [1])**.** *For each d there is $q_0(d)$ such that every Oppermann sub-interval of degree $2d$ over $\mathbb{F}_q$, with $q \geq q_0(d)$, contains an irreducible polynomial in each of the following cases:*

1. *q odd and $d \geq 3$;*
2. *q odd, $d = 2$ and $\gcd(q, 6) = 1$;*
3. *q even and $d \geq 4$;*
4. *q even, $d = 3$, and the class has $p_3 \neq 0$.*

Proof. We check the conditions of [1] with $k = 2d$ and $m = d - 1$; recall that $1 \leq m < k$ is required.

- *q odd.* The only condition is $m \geq 2$ when $p \mid 2d(2d-1)$. It holds for $d \geq 3$. For $d = 2$, $k(k-1) = 12$, so it holds when $\gcd(q, 6) = 1$.
- *q even, $d \geq 4$.* Here $m \geq 3$, so both conditions hold.
- *q even, $d = 3$.* By Lemma 2.2 every member P of an Oppermann class has $p_5 = 0$. In characteristic 2 this gives $P' = p_3 t^2 + p_1$, so $\deg P' = 2$ when $p_3 \neq 0$, and the condition "$m \geq 3$ if $\deg P' \leq 1$" does not apply.

□

The cases left open by the corollary are the following. For $d = 1$ the interval is too short for [1], but Theorem 1.1(A) and, for odd q, the count $(q-1)/2$ of irreducible $t^2 + bt + s_0$ settle it. For $d = 2$ the corollary is superseded: Theorems 1.1(B) and 1.4 give exact counts for every q, including $3 \mid q$, with no threshold. For q even with $d = 3$, the q classes with $p_3 = 0$ (prefixes $(p_5, p_4, p_3) = (0, x, 0)$) remain open. In every case, $q < q_0(d)$ is covered only by computation.

7. Questions

(1) In characteristic 2, are (A)–(C) the only empty Oppermann sub-intervals for all q and d? By Corollary 6.1, for each $d \geq 4$ only finitely many q can carry further empty sub-intervals. For $d = 3$, the classes with $p_3 = 0$ are not covered by [1], but they are nonempty in the computed range $q \leq 16$.

(2) In odd characteristic, is every Oppermann sub-interval nonempty for all q and d? Lemma 2.1 turns this into a statement about all short intervals of length $q^{n/2}$. By Corollary 6.1, for each $d \geq 3$ it fails for at most finitely many q. For $d = 2$ it holds for every odd q (Theorem 1.4). An explicit $q_0(d)$ combined with Theorem 1.2 would settle further small d completely. The trace-form method of Section 4 would need the joint distribution of $\mathrm{Tr}(x^2)$ and $\mathrm{Tr}(x^3)$ on $\ker \mathrm{Tr} \subset \mathbb{F}_{q^6}$ for $d = 3$; it does not reduce to a quadratic form.

(3) Outside the characteristic-2 cases above, is there a constant $c > 0$ with $\min_x \pi(x) \geq c\, q^d/2d$ for all large d?

Data and code

- `hayes_oppermann.py`: Theorems 1.2 and 1.3.
- `verify_pointwise_oppermann.py`: brute-force census, $q^{2d} \leq 2^{26}$.
- `compare_bruteforce.py`: class-by-class cross-check.
- `make_tables.py`, `make_figure.py`: Tables 2, 3 and Figure 1.
- `char2_d2_blocks.py`, `char2_d2_theorem.py`: Theorem 1.1(B).
- `swan_parity.py`: Theorem 1.1(C) and Table 1.
- `odd_d2_formula.py`: brute-force check of Theorem 1.4.
- `hayes_fft_fq.py`, `fqlib.py`, `small_q_gf.py`: taken unchanged from [3].

All scripts are in `code/`; their outputs are in `data/`.

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